

Jonathan Barnes | Applied Data Scientist & Analytics Engineer

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PROFESSIONAL EXPERIENCE

University of Minnesota, Minneapolis, MN

Analytics Engineer | Center for Public Health Systems

(May 2025 - Present)

- Architected a production analytical data platform for all HHS awards, consolidating 400 inconsistent source fields into 4M records and 26 standardized variables enabling global funding analysis that was previously infeasible at scale.
- Designed a fault-tolerant, repeatable ingestion and orchestration layer backed by DuckDB that replaced direct dependence on unreliable, rate-limited government APIs with a persistent analytical store for ad-hoc analysis.
- Developed a custom interface for non-technical users to trigger refreshes and rebuilds, monitor pipeline status, and run constrained ad-hoc queries against the persistent store, enabling self-serve access and reducing routine engineering support.
- Engineered a repeatable, memory-efficient normalization pipeline in Polars processing heterogeneous federal data end-to-end in under 2 minutes using <12GB RAM, reducing memory usage by 80%+ and avoiding high-memory or distributed infrastructure.
- Improved analytical consistency through a repeatable transformation workflow that resolves changing field definitions, missingness, and conflicting values by consolidating 4–8 source fields per variable and validating results against external reference tables to increase accuracy and downstream validity.
- Built a custom Polars-to-Tableau Hyper export layer to reduce analytic file size by over 90%, significantly improving enterprise dashboard performance while preserving full granularity.
- Integrated PHIG grant data with HHS awards to power Tableau dashboards and cross-program visual analysis.

Graduate Teaching Assistant | Division of Biostatistics and Health Data Science

(Aug 2024 - May 2025)

- Taught and evaluated applied statistical analysis in R, improving student code quality and statistical reasoning.

Graduate Research Assistant | Medical School

(Oct 2023 - May 2024)

- Designed custom plasma biomarker phenotypes and gene expression features to enable survival modeling.
- Applied Cox proportional hazards regression to model time to cognitive decline using biomarker phenotypes.

University of St. Thomas, St. Paul and Minneapolis, MN

ITS Support Technician Intern & Senior Student Technician | ITS Support Services

(Aug 2021 - Aug 2023)

- Resolved or consulted on 4,500+ support tickets across hardware, software, networking, and security, serving as the senior escalation layer between the service desk and support teams while shaping the design of standardized process.
- Designed and deployed a Python-based internal platform that standardized high-variance service desk workflows, replacing a legacy tool and becoming the primary interface used by technicians during live support.
- Built guarded, technician-ready interfaces over operating system, directory, and network queries that return consistent diagnostic and identity information and provide access to administrative systems without requiring command-line use.
- Mentored junior student technicians to independently handle complex service desk operations and consultations.
- Imaged and administered software on 1,500+ replacement computers and leveraged retired hardware to develop an in-house virtual lab replacing AWS EC2 instances and reducing infrastructure costs.
- ITS Outstanding Student Award.

EDUCATION

University of Minnesota School of Public Health, Minneapolis, MN

Master of Public Health - Public Health Data Science, GPA 3.73/4.00

Feb 2026

- Capstone – Data Platform Evaluating Federal Spending and Systems.
- President's Student Leadership & Service Award Nominee (2025)

University of St. Thomas, St. Paul, MN

Bachelor of Science - Data Analytics (Biology Domain), Completed while employed full-time

- Minors - Biology (Biochemistry & Computational Biology courses), Applied Statistics.
- Capstone - Development of Obesity and Nutritional Trends Over Time in the United States Using Combined NHANES.

SKILLS

Languages: Python, R, SQL, Bash.

Tools: Shiny, Git, CI/CD, GitHub Actions, Docker, Microsoft Azure, Tableau, ArcGIS.

Competencies: ETL Pipelines, Data Modeling, System Design, Data Acquisition, Longitudinal Data Analysis, Data Visualization, Machine Learning, NLP, LLMs, Experimental Design, Health Informatics, EHR, HIPAA.

Modeling Methods: Regression, Regularization, Tree Models, Supervised Learning, Unsupervised Learning, Dimensionality Reduction, Imputation, Mixed Effects Models, Survival Analysis, Structural Equation Modeling.

Supporting Work

Public Health Federal Funding Evaluation Platform & Dashboard (Grant Supported)

- Developed a modular ingestion and modeling pipeline to process millions of federal public health funding records, spanning prime awards, sub-awards, and derived transactions across inconsistent reporting structures.
- Reconciled fragmented and contradictory HHS data by implementing identity modeling and rules-based normalization to link awards, recipients, and health departments across divergent hierarchies, identifiers, and reporting systems.
- Architected columnar storage and transformation layers using DuckDB and Parquet, reducing data footprint by 90%+ while preserving transaction-level performance, and materialized normalized models into analysis-ready datasets and custom Tableau Hyper extracts for scalable, reproducible analytics.
- Built a GUI-based orchestration layer to manage pipeline execution, validation, and controlled data refresh, supporting non-technical users in generating analysis-ready datasets and dashboards on funding flows, geography, and program-level allocation.

Methods & Tools: Python, Polars, SQL, DuckDB, Apache Parquet, Tableau (Hyper), Data Modeling, Schema Design, Entity Resolution, Data Normalization, Data Pipelines, Data Warehousing, Data Validation, Health Informatics, API Ingestion.

Tech Desk Tools Python

- Designed and deployed a user-facing Python application that serves as the primary interface for Tech Desk workflows, enabling technicians to initiate identity lookup, diagnostics, and institutional tool access through a unified system.
- Implemented identity derivation and system automation logic that abstracts OS and directory complexity into stable, technician-ready outputs, minimizing manual troubleshooting and workflow variability.
- Integrated OS-level commands and directory services with privilege-aware, role-based execution paths to ensure consistent and secure behavior across diverse account, permission, and system states.
- Adopted as the standard Tech Desk application, supporting continuous daily operations since early 2023.
- [Tool Overview](#)

Methods & Tools: Python, PowerShell, Bash, App Development, Active Directory, Azure, Systems Automation, Workflow Orchestration, Error Handling & Defensive Programming, Technical Writing, Packaging and Deployment.

Mapping Misery – Predicting Poor Physical and Mental Health Days in the U.S.

- Built county-level models of mean-centered poor physical and mental health days across ~3,200 U.S. counties using R and a decade of multi-metric predictors.
- Derived county “health-profile” clusters by discretizing 10-year trajectories across 15 indicators into 8 ordered levels (including missingness), learning latent structure with MCA, and clustering with Gaussian Mixture Models.
- Trained elastic net models with 25-fold cross-validated regularization and CART imputation; achieved $R^2 = 0.844$ (RMSE = 0.263, MAE = 0.208) for physical health and $R^2 = 0.645$ (RMSE = 0.375, MAE = 0.295) for mental health on mean-centered outcomes.
- [Project Page](#)

Methods & Tools: MCA, GMM Clustering, Feature Engineering, Longitudinal Data Analysis, Elastic Net Regression, Predictive Modeling, R, Geospatial Visualization, Data Visualization, Cross-Validation, Imputation.

Predicting Child Mortality and Cesarean Delivery Outcomes with Generalized Linear Mixed Models.

- Built Generalized Linear Mixed Models (GLMM) to investigate factors affecting child mortality and cesarean delivery.
- Determined a 1,124.5% increase in the odds of cesarean delivery for hospital births (95% CI: 618.7% - 1,986.3%, $p < 0.001$), a result consistent with logical expectations.
- [Project Page](#)

Skills & Tools: GLMM, GEE, R, Messy Data, Data Visualization, Feature Engineering, Collaboration.

Applied Biostatistics Introduction Modules : Shiny Suite (Grant Supported)

- Designed applied biostatistics modules translating into intuitive frameworks with supporting GitHub hosted Shiny applications.
- [Intuition Lab: Odds, Base Rates Fallacy, Ratio Changes, Gamblers Fallacy](#)
- [Intuition Lab: Sampling, CLT, Regression To The Mean, Spread and Confidence](#)
- [Intuition Lab: Effect Size, Sample Size, Power, Trade-Offs](#)

Methods & Tools: Biostatistics, Statistical Reasoning, Probability, Curriculum Design, Technical Writing, R, Shiny, Data Pipelines, Dashboards, Data Visualization, Data Communication, JavaScript.